UNIVERSITY OF WESTMINSTER

INFORMATICS INSTITUTE OF TECHNOLOGY

TRENDS IN COMPUTER SCIENCE

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QUANTUM COMPUTING

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HISTORY OF COMPUTING



Abacus Beginnings (17th Century):

- Introduction of the abacus as an early tool for simple math.

19th Century Visionaries:

- Highlight the visionary ideas of Charles Babbage and Ada Lovelace.

20th Century Breakthroughs:

- Explore the pivotal role of Alan Turing and others in the development of electronic computers.

WHAT IS VON NEUMANN ARCHITECTURE?

- Von Neumann Architecture is a design framework for computers, proposed by mathematician and physicist John von Neumann in 1945.

- Von Neumann Architecture comprises components that form the backbone of classical computers. [The Central Processing Unit (CPU), Memory (RAM), Control Unit & Arithmetic Logic Unit (ALU)]

- Advantages:
  - Simplicity in design.
  - Flexibility in program storage and execution.



John Von Neumann
(1945)

WHAT IS QUANTUM COMPUTING?



- Quantum Computing is a cutting-edge computational approach based on the principles of quantum mechanics. It utilizes qubits, which can exist in multiple states simultaneously.
- This enables highly efficient parallel processing, enhanced by entanglement, connecting qubits regardless of physical distance.
- Quantum Computing has transformative potential in cryptography, optimization, simulations, and machine learning, positioning it as a frontier in information technology despite being in its early stages of development.

QUANTUM COMPUTING VS VON NEUMANN ARCHITECTURE

	Conventional Computing	Quantum Computing
Memory & Storage	Separates memory and processing.	Allows both storage and processing together.
Data Speed	May have bottlenecks in speed.	Potential to be faster, overcoming limits.
Error Correction	Uses established techniques.	Research Ongoing.
Information Processing	Follows predictable rules.	Bringing new possibilities.

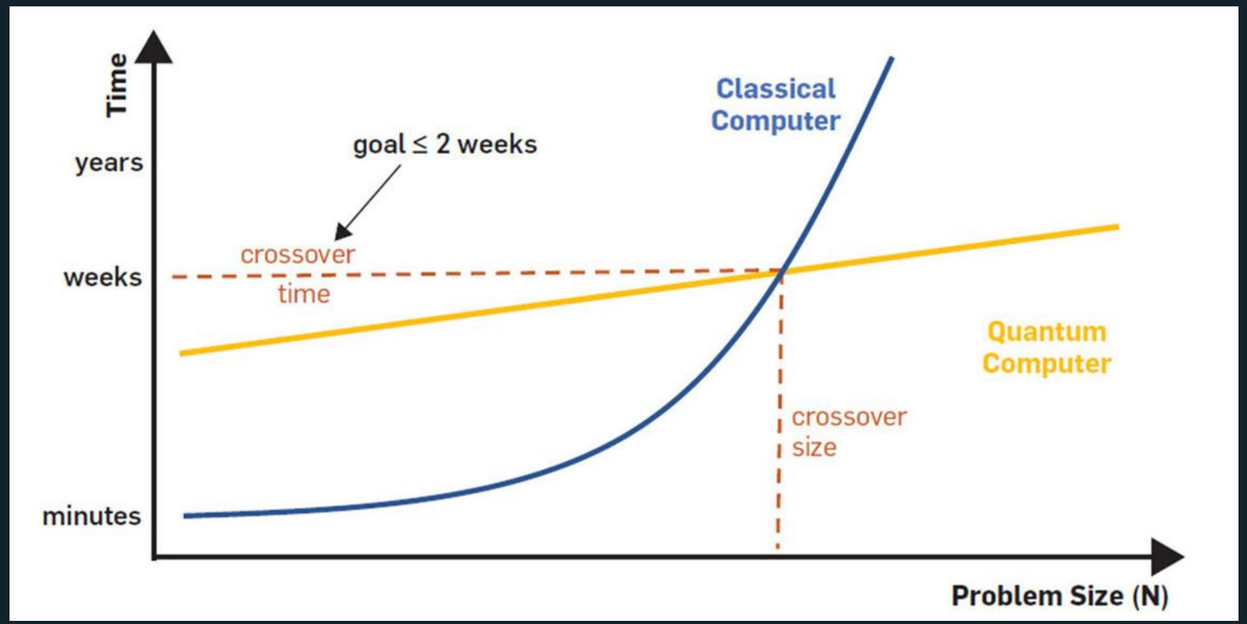
MEMORY AND STORAGE

CONVENTIONAL COMPUTING

In Von Neumann architecture, memory (RAM) and processing (CPU) operate as separate entities, requiring a sequential flow for data transfer and execution.

QUANTUM COMPUTING

Quantum Computing uniquely allows for both storage and processing to occur simultaneously, a distinctive feature stemming from the principle of superposition.



DATA SPEED

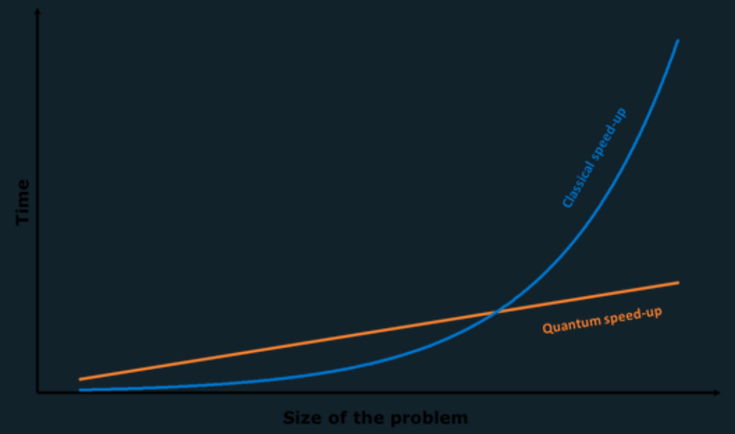
CONVENTIONAL COMPUTING

The potential emergence of bottlenecks in speed is a recognized challenge.

This occurs due to the sequential nature of operations, where data retrieval and processing may face limitations in terms of speed and efficiency.

QUANTUM COMPUTING

Quantum computers have the potential to achieve faster processing speeds, effectively overcoming the limits imposed by the sequential nature of conventional computing.



ERROR CORRECTION

CONVENTIONAL COMPUTING

Error correction relies on well-established techniques.

These proven methods are integral to ensuring the accuracy and reliability of computations.

QUANTUM COMPUTING

The landscape of error correction is currently under active exploration and research.

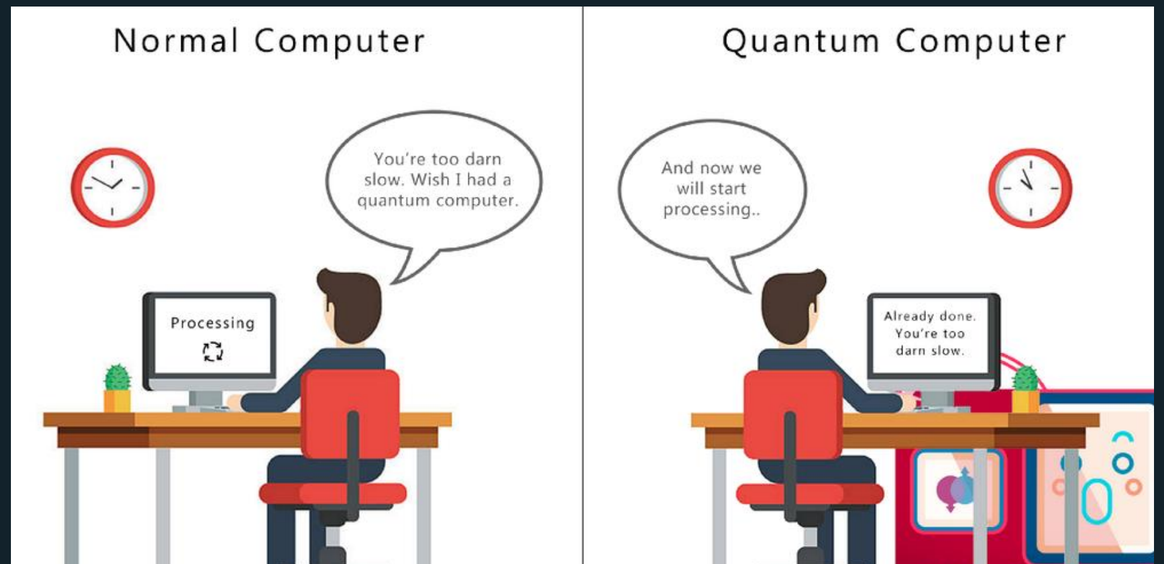
Ongoing efforts are dedicated to developing robust algorithms to address the unique challenges posed by quantum decoherence and errors in quantum systems.

INFORMATION PROCESSING

CONVENTIONAL COMPUTING

Information processing adheres to predictable rules.

This deterministic approach ensures consistent and expected outcomes in data manipulation and computational tasks.



QUANTUM COMPUTING

Quantum Computing introduces a realm of unprecedented possibilities in information processing.

By embracing probabilistic principles and leveraging the unique attributes of qubits, quantum computers offer a departure from the predictability of classical systems, opening avenues for innovative solutions to complex problems.

CONCLUSION

In closing, Quantum Computing propels us into a new era of computational prowess. The paradigm shift from classical bits to quantum qubits, with their unique superposition property, unveils unparalleled potential. As we explore the efficiency in problem-solving and real-world applications, it's evident that Quantum Computing is not just a technological leap but a transformative force. The future promises collaborative integration with classical systems, opening doors to unprecedented computational capabilities. Let's embark on this quantum journey together, where the boundaries of what's possible are redefined. Thank you for joining us today.

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